

Agentic Financial Advisor Bot for Investment Strategies

Author: Kevin Juandi

1. Project Overview

The Agentic Financial Advisor Bot is an AI-powered tool designed to provide intelligent, explainable, and adaptive financial advice for investors. It combines real-time market data processing, predictive modeling, reinforcement learning, and natural language generation to deliver clear and actionable investment strategies. The system enables users to interact with the bot conversationally, receiving customized investment guidance while maintaining transparency and control.

2. Template Used

The project is based on the "Financial Advisor Bot" template. This template focuses on simulating an intelligent agent capable of portfolio management and financial decision-making.

3. Domain and Users

Domain: Financial technology and personal investment advisory.

Users:

- Retail investors seeking low-cost, accessible investment advice.
- Beginner investors overwhelmed by data and lacking financial expertise.
- Users looking for a transparent alternative to black-box robo-advisors.

The project addresses the intersection of personal finance and artificial intelligence, providing an intelligent interface for individuals to manage their investments actively without incurring the high costs of traditional advisors.

4. Design Justification

The project is designed to address three main user challenges:

- **Information Overload:** Users are bombarded with excessive and often conflicting financial information.
- **Advisor Costs:** Traditional financial advisors remain unaffordable for many retail investors.

- **Lack of Adaptability in Tools:** Most existing tools are passive and lack explainability.

The design choices—real-time data integration, predictive modeling, reinforcement learning for strategy selection, and natural language generation—are directly aligned with these user needs. The agentic approach ensures explainability and adaptability, while leveraging existing APIs and LLMs enhances accessibility.

5. Project Structure

The bot is structured as a modular agent with the following architecture:

1. Perception:

- Retrieves and preprocesses live financial data using APIs (e.g., Yahoo Finance, Bloomberg).

2. Reasoning:

- Applies predictive models such as reservoir computing to forecast trends.

3. Action Selection:

- Uses reinforcement learning (via FinRL) to determine optimal portfolio strategies.

4. Natural Language Reporting:

- Converts quantitative decisions into clear, conversational advice using Qwen3.

5. Self-Evaluation:

- Continuously logs predictions and evaluates performance through backtesting.
-

6. Technologies and Methods

- **Reservoir Computing:** Time-series prediction model with low computational cost.
 - **FinRL:** Reinforcement learning framework for financial trading with built in Reinforcement Learning and backtesting feature.
 - **Qwen3:** a family of Large Language Model. For this project, we would be using the small and lightweight variant.
 - **Smolagents:** Lightweight agentic system framework.
 - **Yahoo Finance, Bloomberg, CCXT APIs:** Live financial data.
-

7. Work Plan

The following list outlines the planned phases of the project. The primary goal is to develop a working prototype by Week 12. ***Gantt Chart needed later***

Week Task

- | | |
|-------|---|
| 1 | Ideation and Planning |
| 2-4 | Literature review, finalize tools and technologies |
| 5 | Prototype data ingestion and preprocessing module |
| 6 | Develop and test predictive models |
| 7-8 | Connect LLM (Qwen3) with agentic framework (Smolagents) for natural language output |
| 9 | Integrate FinRL for action strategy generation and backtesting |
| 10–11 | Internal testing, performance tuning, and system integration |
| 12 | Demonstrate working prototype with a single stock (AAPL) |
-

8. Evaluation Plan

The evaluation will address whether the system meets its goals of providing explainable and adaptive investment advice. It includes:

1. Quantitative Evaluation:

- **Backtesting Accuracy:** Comparison of predicted vs. actual returns with historic data.
- **Portfolio Performance:** Benchmarking against passive investment strategies.

2. Qualitative Evaluation:

- **User Feedback:** A/B testing interface clarity and perceived trustworthiness.
- **Transparency Metrics:** Assessment of natural language output quality.

3. Success Criteria:

- Forecast accuracy within 5–10% margin of error over a 3-month simulation.
 - Users report improved understanding and satisfaction with investment decisions.
-

9. Contingency Plan

- If LLM integration fails, fallback to templated explanations based on model output.

- If predictive performance is insufficient, rely on conservative rule-based baselines.
 - If real-time data is inconsistent, use simulated datasets for testing and demonstrations.
-